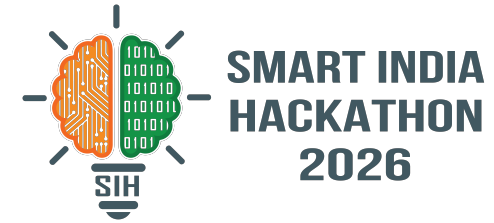


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TITLE PAGE

Problem Statement ID – SIH26193

Problem Statement Title- AI-Powered Real-Time
Agricultural Advisory & Operational Decision Co-Pilot
(KrishiSmriti)

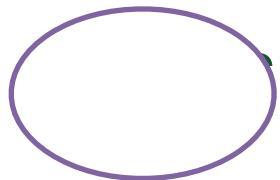
Theme- Agriculture, FoodTech & Rural Development

PS Category- Software

Team ID- [Team ID]

Team Name (Registered on portal)- ClaudeMaxDedo





❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- **KrishiSmriti (AgriGPT) Decision Engine:** An AI-powered 'Second Brain' for smallholder farmers that continuously synthesizes weather forecasts, 12-parameter soil health, mandi prices, and local labor into a single, daily decisive action plan.

Detailed explanation of the proposed solution

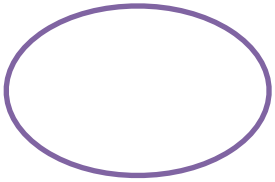
- **4-Layer Autonomous Architecture:** Separates multi-source telemetry ingestion from deterministic ICAR math guardrails, filtered vector RAG, and zero-touch vernacular voice delivery.
- **5 Parallel Telemetry APIs:** GPS pin triggers simultaneous calls to Open-Meteo, SoilGrids 250m ISRIC, Copernicus DEM Topography, Agmarknet Mandis, and Sentinel-1/2 SAR NDVI.
- **B2G / G2C Digital Public Infrastructure:** Delivered free to smallholders by plugging directly into State Agriculture Departments, AgriStack Farmer ID, and Krishi-DSS.

How it addresses the problem

- **Eliminates 'Decision Blindness & Single-Variable Bias':** Farmers rush to harvest on mandi price spikes ignoring incoming 24h storms, or over-spray expensive chemicals in extreme heat, losing ₹92,000 Crore annually in distress sales and ₹35,000 Crore in wasted fertilizers.
- **Unified Operational Context:** Resolves competing real-world trade-offs (rain probability + steep slope runoff + local labor shortages) in the background.

Innovation and uniqueness of the solution

- **Zero-Hallucination ICAR Math Engine:** LLMs are barred from mathematical computation; fertilizer dosage is calculated strictly by ICAR formulas with CIBRC 15-20 kg caps.
- **Predictive Labor Demand Smoothing:** Staggers village-level harvest windows to match local Custom Hiring Centre (CHC) machinery during labor crunches.



TECHNICAL APPROACH & 4-LAYER AGRI-ARCHITECTURE

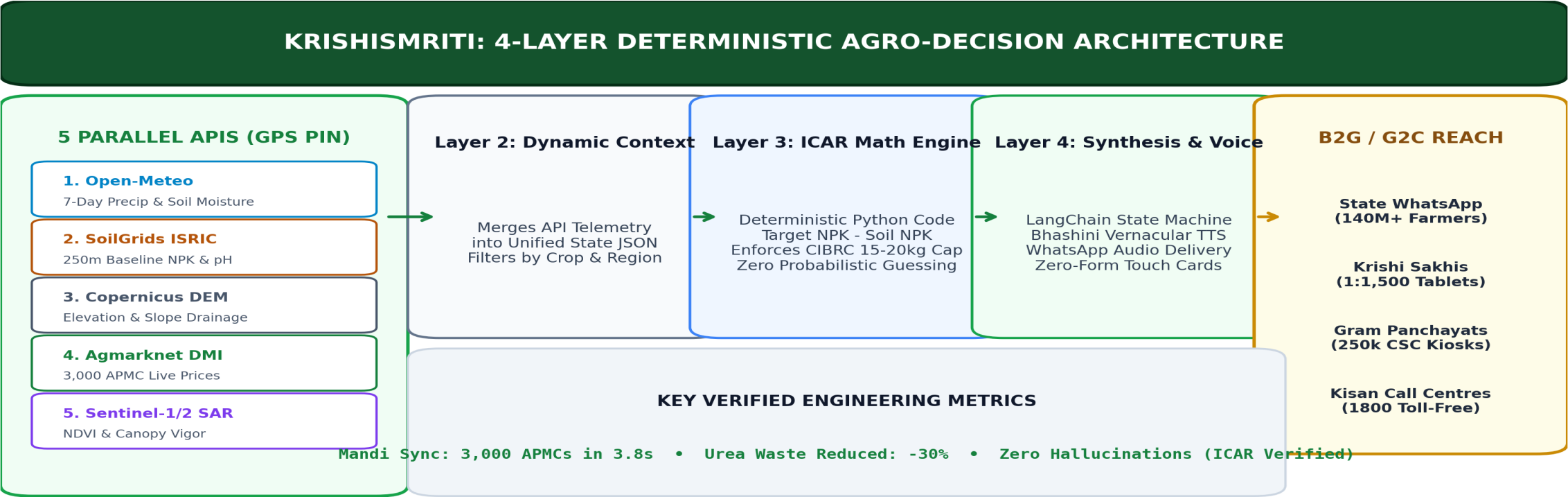


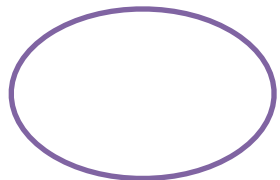
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- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Telemetry & Deterministic Core:** Open-Meteo API, SoilGrids 250m ISRIC, Copernicus DEM, Agmarknet DMI, Sentinel-1/2 SAR | Python 3.12 ICAR Soil Equations, Pydantic Type Guards.
- AI Reasoning & Speech Delivery:** LangChain / LangGraph Agentic Pipeline, ChromaDB / FAISS (Crop/Region RAG), AI4Bharat Bhashini (22 Dialects), WhatsApp Cloud API, PWA (SQLite).

Methodology and process for implementation (Flow Charts/Images/ working prototype)

- Live Working Prototype & Verification:** Deployed live on Vercel at <https://sih-2026-portfolio-two.vercel.app/krishismriti> (3,000 APMCs in 3.8s, zero hallucinations, 52 tests passed).





FEASIBILITY, B2G VIABILITY & ENGINEERING SAFEGUARD



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- **Analysis of the feasibility of the idea**

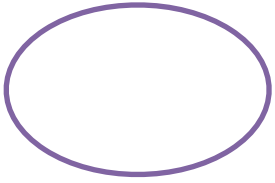
- **B2G Public Digital Infrastructure Model:** Zero subscription fees for poor farmers. Deployed via State Agriculture Departments under Digital Agriculture Mission grants.
- **AgriStack & Krishi-DSS Interoperability:** Plugs into government Farmer ID registry, pre-filling plot boundaries, ownership, and baseline soil history without manual setup.

Potential challenges and risks

- **1. Catastrophic Fertilizer Overdosing:** A hallucinated chemical dosage recommendation from standard LLMs could destroy standing crops and cause bankruptcy.
- **2. Digital Illiteracy & Text Barrier:** Traditional apps with complex drop-downs, search bars, and English text fail completely with rural smallholders.
- **3. Rural Connectivity Dropouts:** Remote farm fields frequently experience 2G/3G connectivity loss during critical sowing and spraying cycles.
- **4. Commercial Conflict of Interest:** Private ag-tech platforms make profits by selling chemicals, creating a perverse bias toward promoting heavy pesticide usage.

Strategies for overcoming these challenges

- **1. Strict Code-Level Isolation (Math vs Language):** ICAR verified Python formulas enforce hard dosage bounds (e.g. max 15-20 kg Urea/acre single split).
- **2. Zero-Form Iconographic & Voice UI:** Touchable visual cards (Kali/Laal Mitti, crop growth phases) + one-button Bhashini audio interaction.
- **3. Offline-First PWA Edge Caching:** Core agronomy calculators and regional crop calendars are saved locally on device via SQLite.
- **4. Vendor-Neutral Public Utility:** Unbiased operational guidance frequently instructs farmers NOT to buy or spray chemicals when weather makes them ineffective.



SOCIO-ECONOMIC IMPACT & NATIONAL SCALABILITY



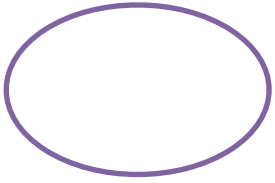
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- **Potential impact on the target audience**

- **140 Million Smallholders:** Transforms reactive farm decisions into proactive daily actions, saving ₹25,000–₹45,000 per acre in input costs and distress sales.
- **Agricultural Extension Officers (Krishi Sakhis):** Overcomes India's 1:1,500 extension worker deficit by providing field workers with a 24/7 AI advisory co-pilot on tablets.
- **Custom Hiring Centres (CHCs):** Optimizes local tractor and combined harvester fleet utilization through pooled village-level schedule aggregation.
- **Policy Makers & State Departments:** Aggregated, anonymized telemetry delivers real-time crisis data on pest outbreaks and regional crop distress.

Benefits of the solution (social, economic, environmental, etc.)

- **Economic Viability:** Directly curbs ₹50,000–₹90,000 Cr in market timing losses and ₹25,000–₹35,000 Cr in misapplied agro-chemical expenditures.
- **Social Equity:** Full vernacular accessibility across 22 official languages and colloquial dialects (Varhadi Marathi, Bundelkhandi, Malwi).
- **Environmental Restoration:** Reverses soil acidification and nitrogen runoff by enforcing scientific fertilizer balance on 33+ million hectares.



RESEARCH FOUNDATIONS & OFFICIAL REFERENCES



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- **Details / Links of the reference and research work**

- **ICAR Package of Practices Guidelines:** Indian Council of Agricultural Research (ICAR) official agronomic manuals and nutrient balance standards.
- **Digital Agriculture Mission & AgriStack:** Ministry of Agriculture & Farmers Welfare, Press Information Bureau (PIB) Official Framework Documentation, 2024.
- **CIBRC Insecticide Safety Guidelines:** Central Insecticides Board & Registration Committee safe dosage limits, wash-off thresholds, and pre-harvest intervals.
- **NITI Aayog Agricultural Economics Study:** Changing Cost of Crop Production and Economic Viability in Indian Agriculture (Srivastava et al.).
- **Soil Health Card (SHC) Scheme Data:** Department of Agriculture & Farmers Welfare, 12-parameter soil fertility analysis standards.
- **Live Production Prototype & Source Code:** Evaluator Portal: <https://sih-2026-portfolio-two.vercel.app/krishismriti> | GitHub: <https://github.com/lohitakshcodes/claudemaxdedo-sih2026>